The Frohmanian Tether: A Machine-Checked Audit of Its Architecture and Obstructions

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Abstract

We audit the claims of the technical compilation *The Frohmanian Symplectic Tether: Architecture, Reduced Jacobi Identity, and the Frohmanian Curve* (Zenodo record 23942466) against a Lean 4/Mathlib formalization. Every claim is classified as *proved*, *conditional* (proved under explicitly stated hypotheses), *refuted* (its negation is machine-checked), or *not formalized*. The library builds without `sorry`, and the 37 key theorems depend only on the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. We report proved and conditional results only as such. The compilation's claims of Jacobi vanishing on the reduced orbit and of energy-budget damping are refuted. Its Clay-standard global regularity claim is **not** established: it holds only conditionally on open hypotheses. This paper makes no Millennium Prize claim.

1 Artifact

Repository: `https://github.com/BenFrohman/frohmanian-tether-verified`. The claim-by-claim map is in `FTVerified/Paper.lean`. Build with `lakebuild` (2684 jobs) and audit with `lakeenvleanscripts/AxiomAudit.lean`. Sources derive from `https://github.com/BenFrohman/frohmanian-symplectic-tether-may-form` at `878b311` and `https://github.com/BenFrohman/frohmanian-tether-geometric` (branch `gateaux-pairing`) at `9e40ae7`. The toolchain is Lean v4.30.0-rc1 with Mathlib.

2 Proved results

- **Locality and nonpositive self-pairing (C1/C3).** `frohmanianTether_self_nonpos`, `tetherKernel_self_nonpos`.
- **Exact degeneracy on kinetic energy (C2).** `frohmanianTether_degenerate`, `tetherKernel_degenerates_on_kinetic_energy`.
- **Nine-term Jacobi expansion.** `jacobiator_nine_terms` and `jacobi_iff_mixed_cancel`: the tethered jacobiator vanishes iff the mixed terms cancel. See also `FieldJacobi.jacT_eq_mixedSum` and `FieldJacobi.jacF_eq_zero`.

3 Conditional results

Each holds only under the hypotheses stated in Lean.

- **Negative feedback with a fixed coefficient (C3).** This is finite-dimensional and assumes the Route-A hypothesis: `frohmanianTether_feedback`, `feedback_fixes_coefficient`, `routeA_determines_kappa`.
- **Uniqueness (C4/C5).** This holds within the minimal admissible class only: `uniqueness_of_minimal_metric_tether`.
- **Chevalley–Eilenberg closure.** $d^2 = 0$ holds when B is a coboundary (`CE.d2_d1`, `CE.cocycle_of_coboundary`). The symmetric tether is *not* a CE cochain (`CE.symPart_not_cochain`).
- **Jacobi in special cases.** It holds for central u (`jacobiator_eq_zero_of_central`) and for the x-only shear class (`FieldJacobi.shearY_jacobi_holds`). `tethered_jacobi_identity` assumes the vanishing.
- **Majorant and quartic absorption.** This holds given the ODE inequality: `comparison_bound`, `layer2_bound`, `Riccati.tethered_ceiling`. Without absorption, Riccati blow-up occurs: `Riccati.riccati_blowup`.
- **Maximum-principle feedback.** These are pointwise lemmas: `viscous_nonpos_at_max`, `periodic_exists_max`.
- **Global regularity.** `frohmanian_symplectic_tether_theorem` and `frohmanian_symplectic_tether_theorem_global` derive smoothness from hypotheses, including a local-continuation hypothesis, that are *open*. They do not resolve the Clay problem.

4 Refuted claims

- **The tether damps the energy budget.** This is refuted. C2 forces $B(M, H) = 0$, so the tether contributes nothing to dH/dt: `ObstructionAudit.flow_contribution_zero_of_C2`, `ObstructionAudit.bilin_kernel_of_nonpos_of_self_zero`, `ObstructionAudit.energy_conservation_not_C2`.
- **Jacobi vanishes on the reduced (divergence-free) orbit.** This is refuted. It fails for the divergence-free field $u = (\sin y, 0, 0)$: `JacobiWitness.tethered_jacobiator_ne_zero`, `FieldJacobi.shearY_jacobi_fails_transverse`, `FieldJacobi.divFree_aZ`. The bracket is metriplectic, not Poisson (`FieldMetriplectic.dissF_nonpos`).
- **The shear example shows the reduced orbit is necessary.** This is partly refuted. The failure for the non-divergence-free shear is correct (`FieldJacobi.shearX_jacobi_fails`, `FieldJacobi.not_divFree_uX`), but divergence-free fields also fail.

5 Open and not formalized

The following have no Lean counterpart and remain open:

- Alain's Curve (§2);

- the Frohmanian Curve $J_{\mathrm{red}} = 0$ (§4), for which the compilation gives no explicit J_{red};
- the symplectic, Melnikov, spectral, and holographic representations (§5);
- the explicit majorant ODE $y' = Cy^2\sqrt{y^3} + \frac{\pi}{4}C_{\mathrm{abs}}$.

The Clay-standard global regularity claim is open.

6 Conclusion

The C1–C3 architecture and the algebraic Jacobi structure are machine-checked. Its two central dynamical claims, reduced-orbit Jacobi and energy damping, are false as stated. Global regularity remains conditional on open hypotheses.

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